Strand 3: Geometry and Measurement

Sub-strand 2: Measurement

Lesson 5 (Requested topic): Parts of a Circle and Circle Measures

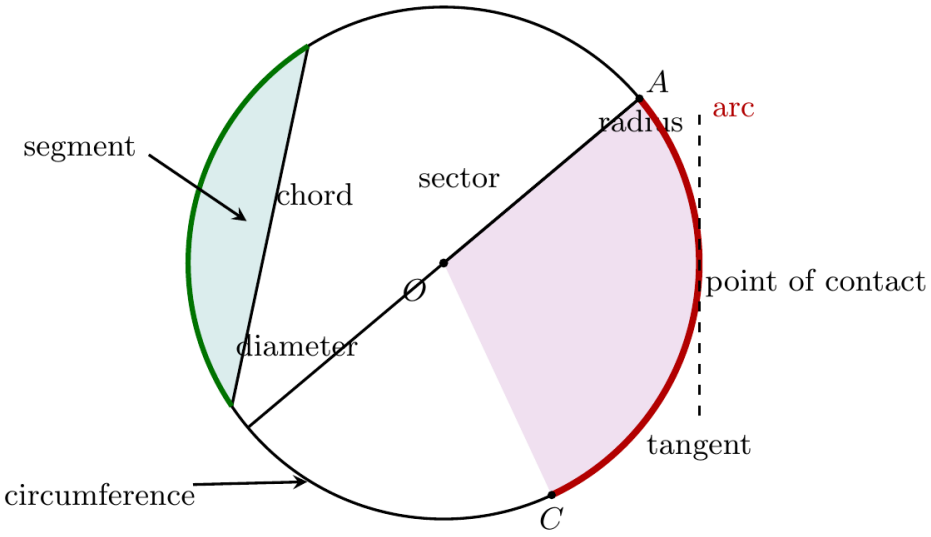
1. Curriculum Standard and Overview

- Strand: 3 — Geometry and Measurement
- Sub-strand: 2 — Measurement
- Content Standards: B7.3.2.1 and B8.3.2.1 (Measurement of circles).
- Indicators: B7.3.2.1.2 (deduce the circumference formula from the diameter relationship), B8.3.2.1.1 (circumference and area of circles); sector measures as B9 application.
- Mock-paper links: Mock P1 Q5 (area 64π → circumference), Mock P2 Q3(d) (semicircle area, r=7). Both are re-taught slowly here with the full part-naming first.

Prerequisites delivered: perimeter & area basics (Lesson 3 of this folder) — because "area of a circle = 154 cm^2" only means something once square units and reverse working are natural to you.

2. Preamble — Naming Every Part First

A circle is the set of all points the same distance from one fixed point, called the centre.



Learn this picture cold. Every part in words:

Part	Meaning	Exam fact
Centre (O)	the fixed middle point	
Radius (r)	centre → any point on the circle	all radii are equal
Diameter (d)	chord through the centre	d = 2r; the longest chord
Circumference (C)	the full boundary length (the "perimeter")	C = π d = 2π r
Chord	a straight line joining two points on the circle	diameter is the longest chord
Arc	a part of the circumference	minor arc vs major arc

Sector	the "pizza slice" between two radii and their arc	like pie chart slices
Segment	the region between a chord and its arc	like a bite of the pizza
Tangent	a line touching the circle at exactly one point (the point of contact)	⊥ to the radius at that point
Semicircle	half a circle (cut along a diameter)	area ½π r^(2); perimeter π r + 2r

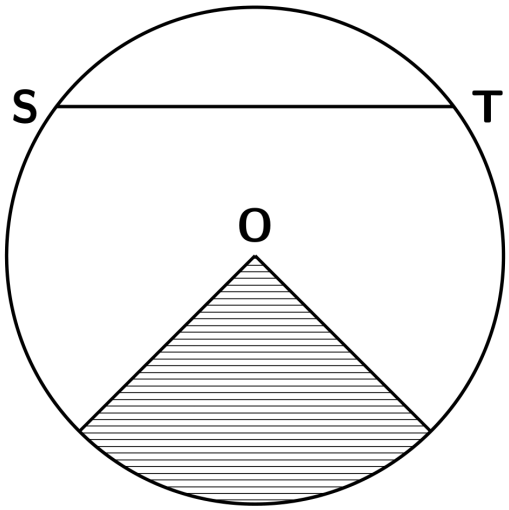


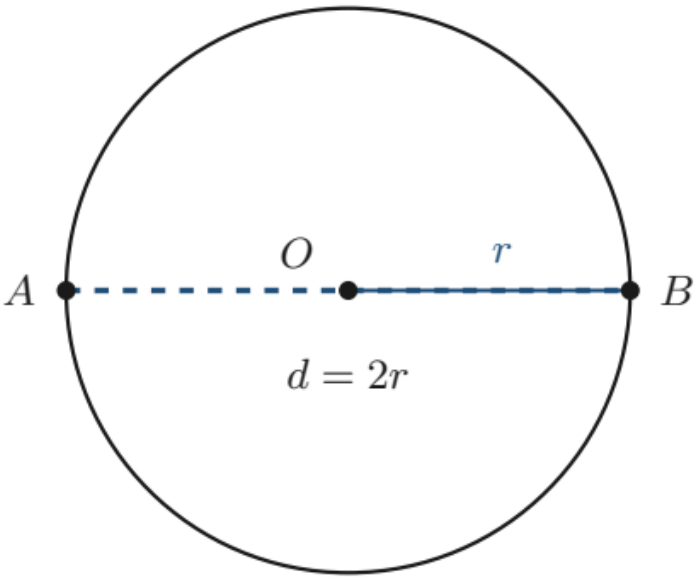
Figure 5 zooms in on the trickiest trio — chord, arc and sector — with the centre shown.

2.1 The number π (pi)

Wrap a string once round any circle and compare it with the diameter: the string is always a bit more than 3 diameters long. That constant is π ≈ 3.14159...

BECE rule: use π = 22/7 when the radius/diameter is a multiple of 7 (the 7s cancel beautifully); otherwise use 3.142. If the paper leaves answers "in terms of π", keep π in.

3. Core Concepts — The Four Measuring Formulas



Measure	Formula	Reverse

Circumference	C = 2π r = π d	r = C/(2π), d = C/π
Circle area	A = π r^(2)	r = √(A/π)
Arc length	l = θ/360 × 2π r	the arc is that fraction of the circumference
Sector area	A_(s) = θ/360 × π r^(2)	same fraction of the area
Sector perimeter	l + 2r	arc + the two radii, not the arc alone

Key-idea (understand, don't memorise): the fraction θ/360 tells you how much of the whole circle you keep. Use it for the arc (part of the boundary) and for the sector area (part of the surface). Sector perimeter adds the two straight radii.

4. Fifteen Fully Worked Examples

Example 1 — Naming parts (authentic BECE definitions)

(a) The distance from the centre to any point of a circle is called what? (b) The line joining two points on the circumference is called what? (c) The longest chord of a circle is the what?

(a) radius; (b) chord; (c) diameter.

Honest note: one banked answer key lists "segment" for (c); the correct mathematics is diameter — a chord through the centre beats every other chord in length.

Example 2 — BECE 2006 (authentic)

Find the circumference of a circle with radius 3.5 cm. {Take π = 22/7}

$$C = 2\pi r = 2 \times \frac{22}{7} \times 3.5 = 2 \times 22 \times 0.5 = 22 \text{ cm}$$

Example 3 — BECE 1991 (authentic, reverse)

The circumference of a circular track is 440 m. Find its diameter. (π = 22/7)

π d = 440 → d = 440 × 7/22 = 140 m.

Example 4 — BECE 2002 (authentic, reverse with decimals)

The circumference of a circular track is 15.4 m. Find the diameter. (π = 22/7)

d = 15.4 × 7/22 = 4.9 m.

Example 5 — BECE 2004 (authentic)

A circle has radius 7 cm; find its area. (π = 22/7)

A = π r^(2) = 22/7 × 49 = 154 cm^2. (Notice: 49 ÷ 7 = 7, the 7s wipe out — that's why 22/7 is given for multiples of 7.)

Example 6 — BECE 2005 (authentic)

Find the area of a circle whose diameter is 28 cm. (π = 22/7)

r = 14; A = 22/7 × 14^(2) = 22/7 × 196 = 22 × 28 = 616 cm^2.

Example 7 — BECE 2010 (authentic, exact-π style)

Find the circumference of a circle whose area is 64π cm^2.

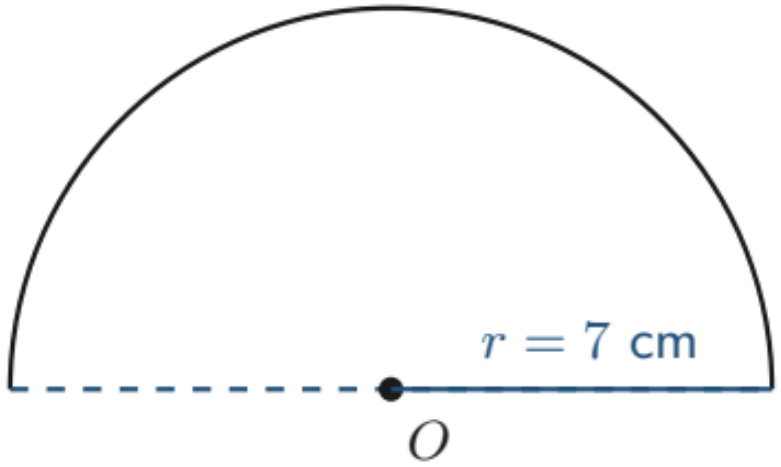
π r^(2) = 64π → r^(2) = 64 → r = 8. C = 2π r = 16π cm. (Watch the options' units: answer is a length, so "16π cm".)

Example 8 — BECE 2022 past style (authentic)

A circular tarpaulin pond has an area of 616 m^2. Find the circumference. (π = 22/7)
π r^(2) = 616 → r^(2) = 196 → r = 14. C = 2 × 22/7 × 14 = 88 m.

Example 9 — BECE 2010-style (authentic)
A circle of diameter 14 cm is cut from a square of side 14 cm. Find the remaining area. (π = 22/7)
Square = 196 cm^2; circle = 22/7 × 49 = 154 cm^2; remaining = 196 - 154 = 42 cm^2.

Example 10 — Semicircle area (mock P2 Q3(d), rebuilt)



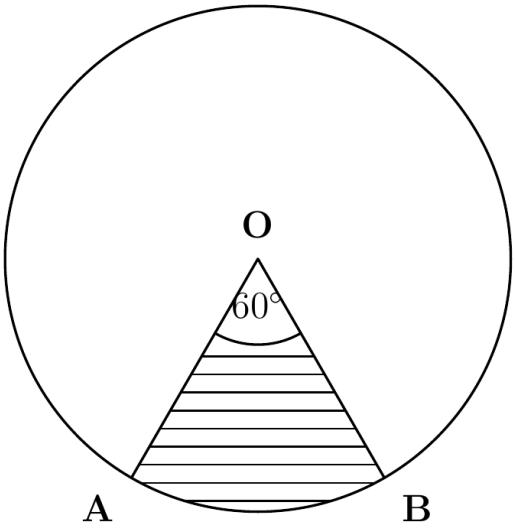
$$\text{Area} = \frac{1}{2}\pi r^2 \qquad \text{perimeter} = \pi r + 2r$$

Find the area of a semicircle with radius 7 cm. (π = 22/7)
Full circle = 154; semicircle = 77 cm^2.

Example 11 — Semicircle perimeter (authentic style)
A semicircle has radius 7 cm. Find its perimeter. (π = 22/7)
Curved part = π r = 22 cm; straight diameter = 14 cm; total = 36 cm.

Example 12 — BECE 1993 (authentic, sector area)
The area of a circle, centre O, is 120 cm^2. angle AOB = 60°. Find the area of the sector AOB.
Fraction = 60/360 = ⅙; sector = ⅙ × 120 = 20 cm^2.

Example 13 — Sector area with π = 22/7 (own, canonical figure)



A sector has radius 21 cm and central angle 120°. Find its area. (π = 22/7)

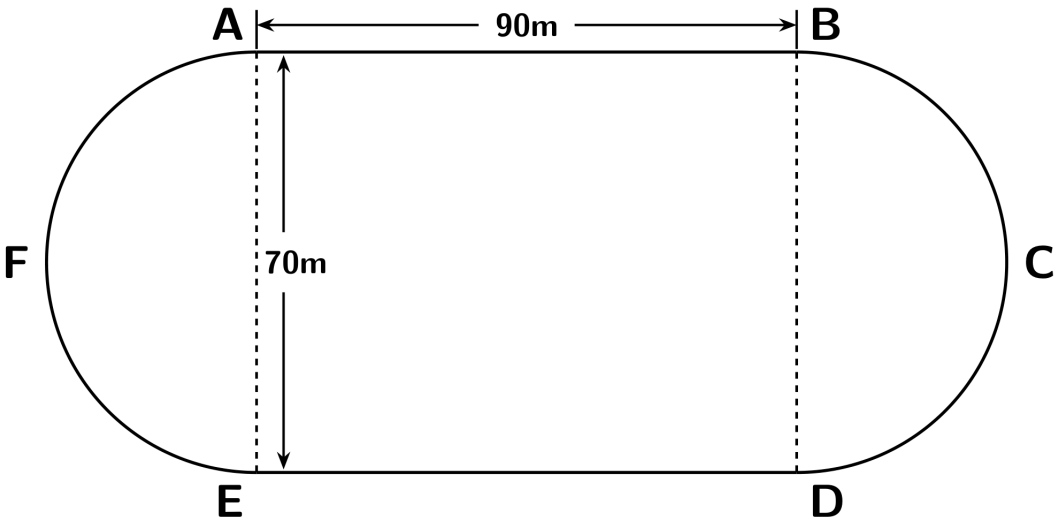
Fraction = 120/360 = ⅓. Area = ⅓ × 22/7 × 21^(2) = ⅓ × 22/7 × 441 = 22/7 × 147 = 462 cm^2.

Example 14 — Arc length (own)

Find the length of the arc of Example 13 (radius 21 cm, angle 120°), and the perimeter of the sector.

C = 2 × 22/7 × 21 = 132 cm; arc = ⅓ × 132 = 44 cm. Sector perimeter = 44 + 2(21) = 86 cm.

Example 15 — Running-track composite (own, canonical figure)



A track consists of a rectangle 80 m by 56 m with two semicircular ends of diameter 56 m. Find (a) the distance a runner travels once round the inner edge, (b) the inner-field area.

(a) Two straights: 2 × 80 = 160 m; two semicircles = one full circle of d = 56: C = π d = 22/7 × 56 = 176 m. Round trip = 336 m. (b) Rectangle 80 × 56 = 4480 m^2; circle = 22/7 × 28^(2) = 2464 m^2; total = 6944 m^2.

5. Exercises — 5 Sections × 5 Questions (No MCQ)

Take π = 22/7 unless told otherwise. Show working + units.

Exercise 1 — Naming and identifying

1. Name the part of a circle: (a) the boundary itself, (b) the "pizza slice", (c) the "bite" between a chord and its arc.
2. The diameter of a circle is 18 cm. What is its radius?
3. A chord passes through the centre of a circle. What is its special name and how does its length compare with every other chord?
4. Which part of a circle touches it at exactly one point, and what is that point called?
5. A sector has central angle 180°. What everyday name does this sector take?

Exercise 2 — Circumference work

1. Find the circumference of a circle of radius 7 cm.
2. Find the circumference of a circle of diameter 21 cm.
3. The circumference of a wheel is 110 cm. Find its radius.
4. A circular pond has circumference 44 m. Find (a) its diameter, (b) its area.
5. A wheel of radius 35 cm rolls one full turn. How far does it travel, in metres?

Exercise 3 — Area work

1. Find the area of a circle of radius 10 cm. (Give your answer to 1 decimal place, $\pi = 3.142$.)
2. Find the area of a circle of diameter 42 cm.
3. The area of a circle is 2464 cm^2. Find its radius.
4. A circle's area equals a square's area of side 22 cm. Show both computations match (square $\pi = 22/7$, $r = 14$).
5. A circle of diameter 21 cm is cut from a rectangle 30 cm by 21 cm. Find the area left over, to 1 d.p. ($\pi = 3.142$)

Exercise 4 — Arcs, sectors, semicircles

1. A sector has radius 12 cm and angle 90°. Find its area and its arc length. ($\pi = 3.142$)
2. A sector has radius 14 cm and angle 135°. Find its area and its perimeter.
3. Find the perimeter of a semicircle of radius 14 cm.
4. The area of a circle is 500 cm^2; a sector subtends 72°. Find the sector's area.
5. The minute hand of a clock is 14 cm long. Find the distance its tip travels in 20 minutes, and the sector "swept" area in those 20 minutes.

Exercise 5 — Spiral review

1. (Area of polygons, previous lesson) A trapezium has parallel sides 15 cm and 9 cm, height 8 cm. Find its area. Then compare: which is bigger, this trapezium or a circle of radius 6 cm ($\pi = 22/7$)?
2. (Triangles) A right triangle has hypotenuse 25 cm and one leg 20 cm. Find the other leg.
3. (Perimeter & area, Lesson 3) A square room of side 7 m has a circular carpet of radius 2.8 m laid in the middle. Find the visible floor area. ($\pi = 3.142$)
4. (Percent) A circular field of radius 70 m has 25% of its area mowed. Find the mowed area. ($\pi = 22/7$)
5. (Algebra) A circle's radius is $(x + 2)$ cm and its area is 154 cm^2. Solve for x.

6. Full Worked Solutions — Exercises

Exercise 1

1. (a) circumference; (b) sector; (c) segment.
2. $r = 9$ cm.
3. Diameter; it is the longest possible chord.
4. The tangent; the touching point is the point of contact.
5. A semicircle (half the circle, 180° of turn).

Exercise 2

1. C = 2 × 22/7 × 7 = 44 cm.
2. C = π d = 22/7 × 21 = 66 cm.
3. r = 110/(2π) = 110 × 7/44 = 17.5 cm.
4. (a) d = 44 × 7/22 = 14 m. (b) A = 22/7 × 49 = 154 m^2.
5. One turn = 2π r = 2 × 22/7 × 35 = 220 cm = 2.2 m.

Exercise 3

1. A = 3.142 × 100 = 314.2 cm^2.
2. r = 21; A = 22/7 × 441 = 1386 cm^2.
3. r^(2) = 2464/22 × 7 = 784 → r = 28 cm.
4. Square = 484 cm^2; circle = 22/7 × 14^(2) = 22/7×196 = 616 cm^2. They do not match — the printed claim is false (the intended pair was square of side 22: area 484, vs circle of r = 14: 616). Say the correct maths and reject the false claim.
5. Rectangle = 630 cm^2; circle r = 10.5: A = 3.142 × 110.25 ≈ 346.35; leftover ≈ 283.7 cm^2 (1 d.p.).

Exercise 4

1. Fraction ¼: area = ¼ × 3.142 × 144 = 113.1 cm^2; arc = ¼ × 2 × 3.142 × 12 = 18.9 cm.
2. Fraction = 135/360 = ⅜: sector = ⅜×22/7×196 = 231 cm^2; arc = ⅜× 2×22/7×14 = ⅜×88 = 33 cm; perimeter = 33 + 28 = 61 cm.
3. π r + 2r = 44 + 28 = 72 cm.
4. Fraction 72/360 = ⅕; sector = 100 cm^2.
5. In 20 min the hand turns 120° (fraction ⅓ of a turn): distance = ⅓ × 2 × 22/7 × 14 = 88/3 ≈ 29.3 cm; swept area = ⅓ × 22/7 × 196 = 154/3 ≈ 51.3 cm^2.

Exercise 5

1. Trapezium = ½(24)(8) = 96 cm^2; circle = 22/7×36 ≈ 113.1 cm^2 → the circle is bigger by about 17.1 cm^2.
2. √(25^(2) - 20^(2)) = √(225) = 15 cm.
3. Room = 49 m^2; carpet = 3.142 × 7.84 ≈ 24.6 m^2; visible ≈ 24.4 m^2.
4. Full = 22/7 × 4900 = 15 400 m^2; mowed = 0.25 × 15 400 = 3850 m^2.
5. 22/7(x+2)^(2) = 154 → (x+2)^(2) = 49 → x + 2 = 7 → x = 5 (radius positive).

7. Memory Card

- d = 2r; C = π d; A = π r^(2); 22/7 for 7-multiples, else 3.142.
- Arc & sector: multiply the whole thing by θ/360; sector perimeter adds 2r.
- Semicircle: area ½π r^(2), perimeter π r + 2r.
- Names: chord (straight, inside), arc (curved, on the boundary), sector (radii + arc), segment (chord + arc), tangent (just touches), diameter (longest chord).
- Reverse: r = √(A/π).

Next: Constructions I — you now own the circle vocabulary that the incircle/circumcircle constructions will need.